

Module 2: Monte Carlo Methods

MA 563 – Statistical Computation and Simulation

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Unit Summary

A Problem That Shows Up Everywhere

- Across science and engineering, we constantly need to evaluate quantities of the form

$$\theta = \int_{\mathcal{X}} h(x) f(x) dx = E_f[h(X)],$$

where f is a probability density (or an un-normalized weight) and h is some function of interest.

- In an ideal world, we would find θ by pen-and-paper calculus or a numerical quadrature rule.
- In the real world, most integrals that actually matter are **high-dimensional, have no closed form, and are too expensive for classical quadrature.**

The central question of Module 2

Can we estimate $\theta = E_f[h(X)]$ *without* ever solving the integral – using nothing but random samples?

Example: A Simple Integral With No Analytic Solution

- Consider the innocent-looking integral

$$\theta = \int_0^1 \frac{\sin x}{x} dx.$$

- This looks like it should have a simple answer – but it **provably has no elementary antiderivative** (a classical result, via Liouville's theorem): no finite combination of polynomials, exponentials, logarithms, and trig functions differentiates to $\sin(x)/x$.
- So there is no formula you can write down and evaluate at $x = 0$ and $x = 1$ – calculus, as usually taught, simply cannot finish this problem.

But Monte Carlo does not care

Write $\theta = E[h(U)]$ with $U \sim \text{Unif}(0, 1)$ and $h(u) = \sin(u)/u$ (defining $h(0) = 1$ by continuity). Draw $U_1, \dots, U_n$ and average $h(U_i)$ – exactly as easy as any other integral, whether or not a closed form exists.

Example: Integrals Famous Enough to Get Their Own Name

- Some integrals are so common, yet so resistant to elementary evaluation, that mathematicians simply **defined a new function** to be the answer:

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx, \quad B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx.$$

- Neither integrand has an elementary antiderivative for general α, β – yet these two integrals appear *everywhere*: they define the Gamma and Beta probability distributions, and normalize half the densities you meet in statistics.
- We only “know” Γ and B because they have been tabulated numerically – not because anyone found a formula.

The honest picture

Even values you take for granted, like $\Gamma(1/2) = \sqrt{\pi}$, come from evaluating the integral cleverly at that *one* special point – for a generic α (e.g. $\Gamma(2.37)$), there is no shortcut at all.

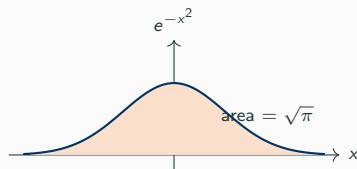
Example: A Famous Integral – Try It Analytically First

- Before turning to genuinely hopeless integrals, try this classic one **by hand**:

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx.$$

- There is **no elementary antiderivative** of e^{-x^2} – you cannot find I by guessing a function to differentiate.
- Yet I has an exact closed form! The classical trick: square it and switch to polar coordinates,

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = \pi, \quad \text{giving } I = \sqrt{\pi}.$$



Classroom challenge

Work out $I = \sqrt{\pi}$ analytically using the squaring-and-polar-coordinates trick above – it is the same polar-coordinates idea behind the Box–Muller transform from Module 1! Then compare it with a Monte Carlo estimate of I once we build the machinery in this module.

Example: Pricing a Path-Dependent Option (Finance)

- An **Asian option** pays off based on the *average* stock price over its life, not just the final price:

$$\text{Payoff} = \max(\bar{S} - K, 0), \quad \bar{S} = \frac{1}{T} \int_0^T S_t dt.$$

- Its fair price is $\theta = e^{-rT} E[\max(\bar{S} - K, 0)]$, an expectation over an entire continuous stock-price path $\{S_t\}_{t \in [0, T]}$.
- There is **no closed-form formula** (unlike the plain vanilla Black–Scholes option) once the payoff depends on the whole path.

Standard practice on every trading floor

Simulate thousands of possible price paths, compute the payoff on each, and average – a direct application of the Monte Carlo principle we are about to formalize.

Example: Bayesian Evidence in Genomics and AI

- In Bayesian inference (genomics: estimating population parameters from SNP data; AI: Bayesian neural networks), the posterior mean of a quantity $h(\theta)$ is

$$E[h(\theta) \mid \text{data}] = \frac{\int h(\theta) L(\text{data} \mid \theta) \pi(\theta) d\theta}{\int L(\text{data} \mid \theta) \pi(\theta) d\theta}.$$

- Both numerator and denominator (the **marginal likelihood** or “evidence”) are integrals over $\theta \in \mathbb{R}^d$, often with d in the hundreds or thousands (e.g. millions of genetic markers, or network weights).
- Classical quadrature is hopeless in such dimensions – this single fact motivates almost all of modern computational statistics.

The Curse of Dimensionality

- A classical quadrature rule (e.g. Simpson's rule) evaluates h on a regular grid.
- To keep the same accuracy in d dimensions, we need roughly k^d grid points, where k is the number of points needed per dimension.
- Even a modest $k = 10$ points per axis becomes 10^{10} evaluations at $d = 10$ – and genomics/finance/AI problems routinely have d in the hundreds.

Preview of the punchline

We will show that a *Monte Carlo* estimator's error shrinks at rate $O(n^{-1/2})$ **regardless of dimension** d . This dimension-free property is precisely why simulation, not quadrature, is the workhorse of high-dimensional computation.

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Unit Summary

Rewriting an Integral as an Expectation

Any integral of the form

$$\theta = \int_{\mathcal{X}} g(x) dx$$

can be rewritten by introducing *any* density f with $f(x) > 0$ wherever $g(x) \neq 0$:

$$\theta = \int_{\mathcal{X}} \frac{g(x)}{f(x)} f(x) dx = E_f \left[\frac{g(X)}{f(X)} \right], \quad X \sim f.$$

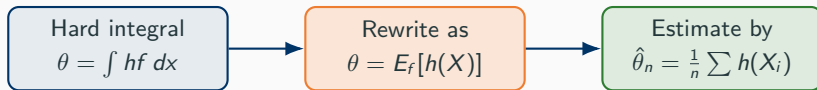
Special case: f already a density

If $\theta = \int h(x)f(x) dx$ with f a genuine probability density, simply take $g(x)/f(x) = h(x)$:

$$\theta = E_f[h(X)].$$

This is the **entire trick** of Monte Carlo: turn “integrate” into “take an expectation”, then estimate the expectation with a sample average.

The Monte Carlo Principle: Integral $\rightarrow$ Expectation $\rightarrow$ Average



- Step 1 (integral $\rightarrow$ expectation) is pure algebra – always possible.
- Step 2 (expectation $\rightarrow$ average) requires **Module 1**: we must be able to draw $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$.
- Everything in Module 2 is about making Step 2 as accurate and efficient as possible.

Crude (Basic) Monte Carlo Estimator

Algorithm: Crude Monte Carlo

1. Generate $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} f$.
2. Compute $h(X_1), \dots, h(X_n)$.
3. Return

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i).$$

Unbiasedness

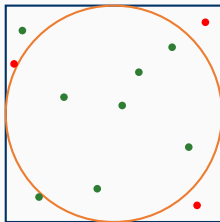
$$E[\hat{\theta}_n] = \frac{1}{n} \sum_{i=1}^n E[h(X_i)] = \frac{1}{n} \cdot n \cdot E_f[h(X)] = \theta.$$

So $\hat{\theta}_n$ is an **unbiased** estimator of θ , for *any* sample size $n \geq 1$.

Classic Example: Estimating π by Throwing Darts

- Imagine throwing a dart uniformly at random into the square $[-1, 1] \times [-1, 1]$.
- Model the landing point as $(X, Y) \sim \text{Unif}([-1, 1] \times [-1, 1])$.
- Define

$$h(X, Y) = \mathbf{1}\{X^2 + Y^2 \leq 1\} = \begin{cases} 1, & \text{if the dart lands inside the unit circle,} \\ 0, & \text{otherwise.} \end{cases}$$



- Since h is an indicator, its expectation is just a probability and under a uniform distribution on the square:

$$E[h(X, Y)] = P(X^2 + Y^2 \leq 1)$$

- Therefore $\pi = 4 E[h(X, Y)]$, and the crude MC estimator is

$$\hat{\pi} = 4 \cdot \frac{1}{n} \sum_{i=1}^n h(X_i, Y_i) = 4 \times (\text{fraction of darts inside the circle}).$$

This is the oldest and most intuitive illustration of the Monte Carlo principle – and it is literally an integral (area) turned into a sample proportion.

Example: An Integral with No Elementary Antiderivative

Target

$$\theta = \int_0^1 e^{-x^2} dx.$$

The integrand e^{-x^2} has **no elementary antiderivative** (it defines the error function, erf) – yet Monte Carlo handles it exactly as easily as any other bounded function.

Crude Monte Carlo solution

Write $\theta = \int_0^1 e^{-x^2} \cdot 1 dx = E[h(U)]$ with $U \sim \text{Unif}(0, 1)$ and $h(u) = e^{-u^2}$.

1. Draw $U_1, \dots, U_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$.
2. Return $\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n e^{-U_i^2}$.

The true value is $\theta \approx 0.7468$; with $n = 10,000$ draws, $\hat{\theta}_n$ typically lands within about ± 0.005 of this. **We will see this in our lab.**

Example: A Genuinely High-Dimensional Integral

Target (engineering / physics flavour)

$$\theta = \int_{[0,1]^{10}} \cos(x_1 + x_2 + \cdots + x_{10}) \, dx_1 \cdots dx_{10}.$$

A 10-dimensional integral over the unit hypercube – exactly the kind of object that appears in uncertainty quantification for engineering systems with many uncertain inputs.

- Classical quadrature with even $k = 5$ points per axis needs $5^{10} \approx 9.8$ million evaluations.
- Crude Monte Carlo: draw $\mathbf{U}_i = (U_{i1}, \dots, U_{i10}) \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]^{10}$ and average $\cos(U_{i1} + \cdots + U_{i10})$ – the *same* $n = 10,000$ draws that worked in 1-D still give a usable answer.
- This dimension-free behaviour is the reason Monte Carlo dominates high-dimensional integration.

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Unit Summary

Why $\hat{\theta}_n$ Actually Works: The LLN

Theorem (Strong Law of Large Numbers)

If $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} f$ and $E_f|h(X)| < \infty$, then

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i) \xrightarrow{\text{a.s.}} \theta = E_f[h(X)] \quad \text{as } n \rightarrow \infty.$$

- This is what justifies calling $\hat{\theta}_n$ a **consistent** estimator: more simulations $\Rightarrow$ the answer gets arbitrarily close to the truth, with probability 1.
- Unbiasedness (every finite n) and consistency (as $n \rightarrow \infty$) together are the two pillars that make crude Monte Carlo trustworthy.
- The LLN says *nothing* about the finite-sample error, only that it eventually vanishes – for that we need the CLT.

Quantifying Stochastic Error

Suppose we estimate $\theta = E_f[h(X)]$ by the sample mean

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i), \quad X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f.$$

- **Central Limit Theorem:** if $\sigma^2 = \text{Var}_f[h(X)] < \infty$, then for large n ,

$$\hat{\theta}_n \stackrel{\text{approx}}{\sim} N\left(\theta, \frac{\sigma^2}{n}\right).$$

- **Monte Carlo Standard Error:**

$$\text{SE}(\hat{\theta}_n) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}},$$

where s is the sample standard deviation of $h(X_1), \dots, h(X_n)$.

Approximate $(1 - \alpha)$ confidence interval

$$\hat{\theta}_n \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}.$$

The $1/\sqrt{n}$ Rate – Blessing and Curse

Key fact

Monte Carlo error shrinks at rate $O(1/\sqrt{n})$, **regardless of the dimension** of X .

- **Blessing:** unlike classical quadrature (whose cost explodes with dimension – the “curse of dimensionality” from the previous section), Monte Carlo’s convergence rate is *dimension-free*. This is *why* simulation is indispensable in the rendering, finance, and Bayesian examples we opened with.
- **Curse:** to halve the standard error, you must **quadruple** the number of simulations ($n \rightarrow 4n$ gives $SE \rightarrow SE/2$) – convergence is comparatively slow, and each extra digit of accuracy is expensive.

The lever we actually control

Since $SE = \sigma/\sqrt{n}$, there are only two ways to shrink it: increase n (expensive), or **reduce** σ^2 by sampling more cleverly. The rest of this module is devoted to the second option – importance sampling and variance reduction.

Reporting Simulation Results Honestly

Good practice

Never report a Monte Carlo estimate as a bare number. Always accompany it with a measure of its uncertainty:

$$\hat{\theta}_n \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \quad (\text{an approximate } (1 - \alpha) \text{ confidence interval}).$$

- Run the simulation with several different seeds and check that estimates are consistent – this helps distinguish stochastic error from a coding bug or a flawed generator (Module 1!).
- Increasing n should shrink the interval at the theoretical $1/\sqrt{n}$ rate; if it does not, suspect a bug or an $h(X)$ with infinite variance.
- We return to this with concrete diagnostics at the end of the module.

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Special Case: Estimating a Probability

- Suppose $X \sim f$ and we want the probability that X lands in some event/region A .
- We will simulate $X_1, \dots, X_n$ **i.i.d. from** f (i.e., from the distribution that defines θ).
- Take $h(x) = \mathbf{1}\{x \in A\}$. Then

$$\theta = E_f[\mathbf{1}\{X \in A\}] = P(X \in A).$$

- Crude Monte Carlo estimator: $\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \in A\}$ – simply the **sample proportion** of draws landing in A .
- Since $\mathbf{1}\{X \in A\}$ is Bernoulli(θ), its variance is $\theta(1 - \theta)$, so

$$\text{SE}(\hat{\theta}_n) = \sqrt{\frac{\theta(1 - \theta)}{n}}.$$

When Crude Monte Carlo Struggles: Rare Events

Motivating example

A structural engineer needs $\theta = P(X > 6)$ where $X \sim N(0, 1)$ represents a (standardized) stress level, and $\{X > 6\}$ is catastrophic structural failure. The true value is $\theta \approx 9.87 \times 10^{-10}$.

- Crude MC: draw $X_i \sim N(0, 1)$, estimate $\hat{\theta}_n = \frac{1}{n} \sum \mathbf{1}\{X_i > 6\}$.
- Relative standard error $\frac{\text{SE}(\hat{\theta}_n)}{\theta} \approx \frac{1}{\sqrt{n\theta}}$ – to get even a modestly reliable relative error, we would need n in the **billions**.
- In practice: with $n = 10^6$ draws, we expect to see *zero* exceedances at all – the estimator returns $\hat{\theta}_n = 0$, which is useless.

The lesson

Crude Monte Carlo wastes almost all its draws on the “uninteresting” region and almost never visits the region that matters. This is precisely analogous to the naive-rejection failure we saw for truncated Normal sampling in Module 1 – and it motivates the next section.

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- Motivation and the Change-of-Measure Idea

- Simple (Unbiased) Importance Sampling

- Variance and the Optimal Proposal

- Worked Example: Rescuing the Rare-Event Problem

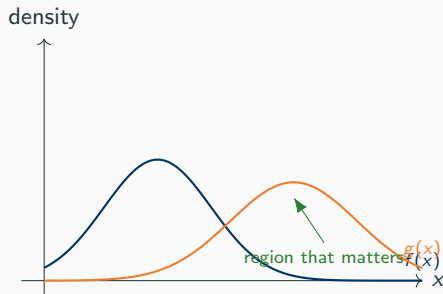
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The Idea: Sample Where It Matters

- Crude MC draws from f itself, so it rarely visits regions where $h(x)f(x)$ is large but $f(x)$ is small (e.g. the tail in the rare-event example).
- **Idea:** draw instead from a cleverly chosen **importance/proposal** density g that puts more mass where $h(x)f(x)$ matters, then **reweight** to correct for the change.



This exactly mirrors the acceptance–rejection idea from Module 1 (choose a helpful “envelope”), except now we *keep every draw* and correct with a weight, rather than throwing draws away.

The Importance Sampling Identity

Theorem (Importance Sampling)

Let g be any density with $g(x) > 0$ whenever $h(x)f(x) \neq 0$. Then

$$\theta = E_f[h(X)] = \int h(x)f(x) dx = \int h(x) \frac{f(x)}{g(x)} g(x) dx = E_g \left[h(X) \frac{f(X)}{g(X)} \right].$$

Algorithm: Simple Importance Sampling

1. Generate $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} g$.
2. Compute **importance weights** $w_i = \frac{f(X_i)}{g(X_i)}$.
3. Return

$$\hat{\theta}_n^{\text{IS}} = \frac{1}{n} \sum_{i=1}^n w_i h(X_i).$$

$\hat{\theta}_n^{\text{IS}}$ is **unbiased** for θ , for any valid choice of g – exactly like crude MC (which is the special case $g = f$, giving all weights $w_i \equiv 1$).

How to Choose the Proposal g ?

- The variance of the importance sampling estimator is $\text{Var}_g[w(X)h(X)]/n$, where $w(x) = f(x)/g(x)$. A poor g can **inflate** variance – even to infinity if g has lighter tails than $f \cdot h$.
- Put more mass of g where $|h(x)|f(x)$ is large – e.g. shift g toward the tail for rare-event problems, exactly as we shifted the envelope for truncated-Normal rejection sampling in Module 1.
- **Heavy tails matter:** if g 's tail is lighter than f 's, the weight $w(x) = f(x)/g(x)$ can blow up, giving a handful of enormous weights that dominate the estimator and destroy its variance – a common and dangerous pitfall.
- **Rule of thumb:** prefer g with tails at least as heavy as f ; check that $\text{Var}_g[w(X)h(X)] < \infty$ before trusting the estimator.

Practical diagnostic

Look at the distribution of the weights $w_1, \dots, w_n$ themselves – if a tiny number of draws carry almost all the weight, the estimator is unreliable no matter how large n is. (We formalize this with **effective sample size** later in the module.)

Rare Event via Importance Sampling

Recall: $\theta = P(X > 6)$, $X \sim N(0, 1)$, $\theta \approx 9.87 \times 10^{-10}$; crude MC needed billions of draws.

Importance sampling fix: shift the proposal into the tail

Take $g(x) = N(6, 1)$ density (i.e. shift the mean to the boundary of the rare event). Then

$$w(x) = \frac{f(x)}{g(x)} = \frac{\varphi(x)}{\varphi(x-6)} = \exp(-6x + 18),$$

where φ is the standard Normal density.

Algorithm

1. Draw $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(6, 1)$ – now roughly *half* the draws exceed 6!
2. $\hat{\theta}_n^{\text{IS}} = \frac{1}{n} \sum_i \exp(-6X_i + 18) \mathbf{1}\{X_i > 6\}.$

With just $n = 10,000$ draws from the shifted proposal, the estimate is accurate to several significant figures – a dramatic improvement over crude MC's complete failure.

When f Is Known Only Up to a Constant

- In Bayesian problems (recall the genomics/AI evidence example), the target is often $f(x) = \tilde{f}(x)/c$ where $\tilde{f}$ is computable but the normalizing constant $c = \int \tilde{f}$ is **unknown**.
- Simple importance sampling needs the exact ratio $f(x)/g(x)$ – impossible if c is unknown.
- **Fix:** form weights from the un-normalized target, $\tilde{w}_i = \tilde{f}(X_i)/g(X_i)$, and **self-normalize**.

Algorithm: Weighted (Self-Normalized) Importance Sampling

1. Generate $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} g$.
2. Compute $\tilde{w}_i = \tilde{f}(X_i)/g(X_i)$.
3. Return

$$\hat{\theta}_n^{\text{WIS}} = \frac{\sum_{i=1}^n \tilde{w}_i h(X_i)}{\sum_{i=1}^n \tilde{w}_i}.$$

Sketch: Consistency of the Weighted Estimator

- Write the estimator as a ratio of sample means:

$$\hat{\theta}_n^{\text{WIS}} = \frac{\frac{1}{n} \sum_{i=1}^n \tilde{w}_i h(X_i)}{\frac{1}{n} \sum_{i=1}^n \tilde{w}_i} =: \frac{A_n}{B_n}, \quad X_i \stackrel{\text{iid}}{\sim} g.$$

- By the LLN,

$$A_n \rightarrow E_g[\tilde{w}(X)h(X)], \quad B_n \rightarrow E_g[\tilde{w}(X)] \quad \text{a.s.}$$

- Compute the limits using $\tilde{w}(x) = \tilde{f}(x)/g(x)$:

$$E_g[\tilde{w}(X)h(X)] = \int \tilde{f}(x)h(x) dx = c E_f[h(X)] = c\theta, \quad E_g[\tilde{w}(X)] = \int \tilde{f}(x) dx = c.$$

- Therefore $\hat{\theta}_n^{\text{WIS}} = A_n/B_n \rightarrow (c\theta)/c = \theta$ (consistency).

Sketch: (Non-)Unbiasedness of the Weighted Estimator

- Even though $E[A_n] = c\theta$ and $E[B_n] = c$, for large n ; and in general

$$E\left[\frac{A_n}{B_n}\right] \neq \frac{E[A_n]}{E[B_n]}.$$

- **Intuition:** A_n/B_n is a **ratio of random variables**; taking expectations does not “pass through” the ratio.
- More formally, with random B_n in the denominator, $E[A_n/B_n]$ typically differs from θ (finite- n bias).
- The good news: the bias usually decays like with large n .

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Variance Reduction Ideas

- The General Idea

- Antithetic Variates

- Control Variates

- Stratified Sampling

- Comparing the Variance Reduction Ideas

Practical Monte Carlo Diagnostics

Recall the lever

$SE(\hat{\theta}_n) = \sigma/\sqrt{n}$: the only way to improve precision *without* more computation is to reduce $\sigma^2 = \text{Var}[h(X)]$ (or its importance-sampling analogue).

Three classical, basic-level variance reduction ideas – each reduces variance using the *same* number of function evaluations n , essentially “for free”:

1. **Antithetic variates** – introduce negative correlation between paired draws.
2. **Control variates** – exploit a correlated quantity whose expectation is already known.
3. **Stratified sampling** – force samples to spread evenly across the domain.

Antithetic Variates: Idea and Algorithm

- For $U \sim \text{Unif}(0, 1)$, note $1 - U \sim \text{Unif}(0, 1)$ too, and $U, 1 - U$ are **negatively correlated**.
- If h is monotonic, $h(U)$ and $h(1 - U)$ are also negatively correlated.
- Pairing them and averaging can **reduce variance** relative to two independent draws.



Algorithm: Antithetic Variates

1. Draw $U_1, \dots, U_{n/2} \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$.
2. Form the n paired values $h(U_i)$ and $h(1 - U_i)$ for each i .
3. Return $\hat{\theta}_n^{\text{AV}} = \frac{1}{n} \sum_{i=1}^{n/2} [h(U_i) + h(1 - U_i)]$.

Antithetic Variates: Why It Works

Variance comparison

Let $Y_i = \frac{h(U_i) + h(1 - U_i)}{2}$. Then

$$\text{Var}(Y_i) = \frac{1}{4} \left[\text{Var}(h(U)) + \text{Var}(h(1 - U)) + 2 \text{Cov}(h(U), h(1 - U)) \right].$$

If h is monotonic, $\text{Cov}(h(U), h(1 - U)) < 0$, so pairing **strictly reduces** variance compared to $n/2$ independent pairs of ordinary crude draws.

- **Finance application:** simulate an option payoff on both a price path driven by Z_i and the “mirror” path driven by $-Z_i$ – a standard variance-reduction trick for Asian and other path-dependent options.
- Cheap to implement (no extra function evaluations needed if we already need n total draws) but the benefit depends on h being reasonably monotonic.

Control Variates: Idea and Algorithm

- Suppose there is a function $c(X)$, **correlated** with $h(X)$, whose expectation $\mu_c = E[c(X)]$ is **known exactly**.
- Define the **controlled estimator**

$$h_\beta(X) = h(X) - \beta(c(X) - \mu_c).$$

- Since $E[c(X) - \mu_c] = 0$, we still have $E[h_\beta(X)] = \theta$ for *any* β – unbiased for every choice.

$\beta^* = \text{Cov}(h(X), c(X))/\text{Var}(c(X))$ is optimal

Use $\text{Var}(h - \beta(c - \mu_c)) = \text{Var}(h) + \beta^2 \text{Var}(c) - 2\beta \text{Cov}(h, c)$. This is a quadratic in β ; differentiate and set to zero: $2\beta \text{Var}(c) - 2\text{Cov}(h, c) = 0 \Rightarrow \beta^* = \text{Cov}(h, c)/\text{Var}(c)$, giving the minimum variance.

Algorithm: Control Variates

1. Draw $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$; estimate $\hat{\beta} = \widehat{\text{Cov}}(h(X), c(X))/\widehat{\text{Var}}(c(X))$ (optimal choice).
2. Return $\hat{\theta}_n^{\text{CV}} = \frac{1}{n} \sum_{i=1}^n [h(X_i) - \hat{\beta}(c(X_i) - \mu_c)]$.

Control Variates: Variance Reduction and an Example

Optimal variance reduction

With $\beta^* = \text{Cov}(h(X), c(X)) / \text{Var}(c(X))$,

$$\text{Var}[h_{\beta^*}(X)] = \text{Var}[h(X)] (1 - \rho^2),$$

where $\rho = \frac{\text{Cov}(h(X), c(X))}{\sqrt{\text{Var}(h(X)) \text{Var}(c(X))}}$. *Derivation:*

$$\text{Var}(h_\beta) = \text{Var}(h - \beta(c - \mu_c)) = \text{Var}(h) + \beta^2 \text{Var}(c) - 2\beta \text{Cov}(h, c).$$

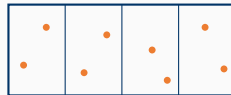
Plugging in β^* gives

$$\text{Var}(h_{\beta^*}) = \text{Var}(h) - \frac{\text{Cov}(h, c)^2}{\text{Var}(c)} = \text{Var}(h) (1 - \rho^2).$$

Since $0 \leq \rho^2 \leq 1$, we have $\text{Var}(h_{\beta^*}) \leq \text{Var}(h)$, with **strict** reduction when $|\rho| > 0$.

Stratified Sampling: Idea and Algorithm

- Crude MC can, by bad luck, cluster many draws in one part of the domain and leave another part sparsely covered.
- **Idea:** partition the domain into K strata of known probability p_k , draw a fixed number from *each* stratum, and combine.



one draw per stratum

Algorithm: Stratified Sampling

1. Partition $\mathcal{X}$ into K disjoint strata with known probabilities $p_1, \dots, p_K$ ($\sum p_k = 1$).
2. In stratum k , draw n_k points from f restricted to that stratum; compute the local mean $\bar{h}_k$.

3. Return $\hat{\theta}_n^{ST} = \sum_{k=1}^K p_k \bar{h}_k$.

Stratified Sampling: Why It Helps

Variance identity

Writing $\sigma_k^2 = \text{Var}(h(X) \mid X \in \text{stratum } k)$ and proportional allocation $n_k = n p_k$,

$$\text{Var}[\hat{\theta}_n^{\text{ST}}] = \frac{1}{n} \sum_{k=1}^K p_k \sigma_k^2 \leq \frac{1}{n} \text{Var}[h(X)] = \text{Var}[\hat{\theta}_n^{\text{crude}}].$$

Stratification **can only help** (or leave variance unchanged) because it removes the “between-stratum” component of variance.

- **Climate application:** when estimating expected regional temperature anomaly under uncertain climate sensitivity, stratifying draws by decade or emissions scenario guarantees every scenario is represented, rather than leaving it to chance.
- Works best when the domain can be naturally partitioned into strata with genuinely different behaviour of h .

Which Technique When?

Technique	Needs	Best suited for
Antithetic variates	Monotonic h	Cheap, quick win; path simulation
Control variates	Correlated $c(X)$ with known $E[c(X)]$	A related, easier problem is solvable
Stratified sampling	A natural partition of the domain	Domain has distinct regimes/regions
Importance sampling	A good proposal g	Rare events, heavy tails, un-normalized f

These are not mutually exclusive

In practice, serious Monte Carlo pipelines (option pricing engines, rendering software, Bayesian samplers) routinely **combine** several of these: e.g. importance sampling *plus* antithetic pairs *plus* a control variate, all at once.

Motivation: The Integrals We Cannot Solve

The Monte Carlo Principle

Law of Large Numbers and Monte Carlo Error

Crude Monte Carlo in Practice

Importance Sampling

Variance Reduction Ideas

Practical Monte Carlo Diagnostics

Trace Plots and Running Means

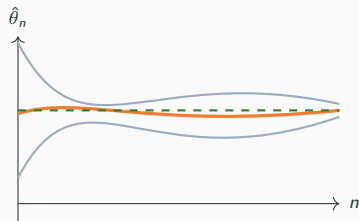
Effective Sample Size for Importance Sampling

Honest Reporting, Revisited

Unit Summary

Diagnostic 1: The Running-Mean Trace Plot

- Plot $\hat{\theta}_k = \frac{1}{k} \sum_{i=1}^k h(X_i)$ against k , alongside the approximate ± 2 SE band.
- A well-behaved estimator: the trace should settle down and the band should visibly narrow like $1/\sqrt{k}$.
- Wild, non-shrinking swings suggest infinite (or very large) variance – a red flag, especially common with poorly chosen importance sampling proposals.



Diagnostic 2: Effective Sample Size (ESS)

- In importance sampling, some draws matter far more than others – a highly **uneven** weight distribution means we effectively have far fewer than n “useful” draws.
- A widely used diagnostic (Kong, 1992):

$$\text{ESS} = \frac{(\sum_{i=1}^n w_i)^2}{\sum_{i=1}^n w_i^2}, \quad 1 \leq \text{ESS} \leq n.$$

- $\text{ESS} \approx n$: weights are roughly equal, close to crude MC efficiency.
- $\text{ESS} \ll n$: a handful of draws dominate – the estimator is effectively based on very few independent pieces of information, however large n nominally is.

Rule of thumb

If ESS is only a small fraction of n (e.g. $\text{ESS}/n < 0.1$), reconsider the proposal g – it likely has tails that are too light relative to the target.

Putting the Diagnostics Together

A responsible Monte Carlo workflow

1. Report $\hat{\theta}_n$ together with its standard error (or, for importance sampling, the ESS-adjusted standard error).
2. Inspect a running-mean trace plot – confirm the estimate has stabilized and the error band is shrinking as expected.
3. For importance sampling, check the ESS and the weight distribution – rule out a single dominant draw.
4. Re-run with a different seed (Module 1!) and confirm consistency.

Why this matters

A single Monte Carlo number with no diagnostics is not a scientific result – it is a guess dressed up in decimal places. These checks are what separate a trustworthy simulation study (in finance, genomics, graphics, or anywhere else) from a silently broken one.

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Unit Summary

Putting It All Together

Method	Core idea	Best suited for
Crude Monte Carlo	Average $h(X_i)$, $X_i \sim f$	Baseline; well-behaved, non-rare targets
Importance sampling	Sample from g , reweight by f/g	Rare events, heavy tails, un-normalized f
Antithetic / control / stratified	Correlate or organize draws	Cheap variance reduction, same n

The Big Picture

- Any hard integral $\theta = \int hf \, dx$ can be rewritten as an expectation $E_f[h(X)]$ and estimated by a sample average – **dimension-free** error at rate $O(1/\sqrt{n})$.
- When f itself is hard to sample, or the region that matters is rare, **importance sampling** reweights draws from an easier proposal g .
- Variance reduction** techniques squeeze more precision out of the same computational budget.
- Always **diagnose** the result: standard error, trace plots, effective sample size.